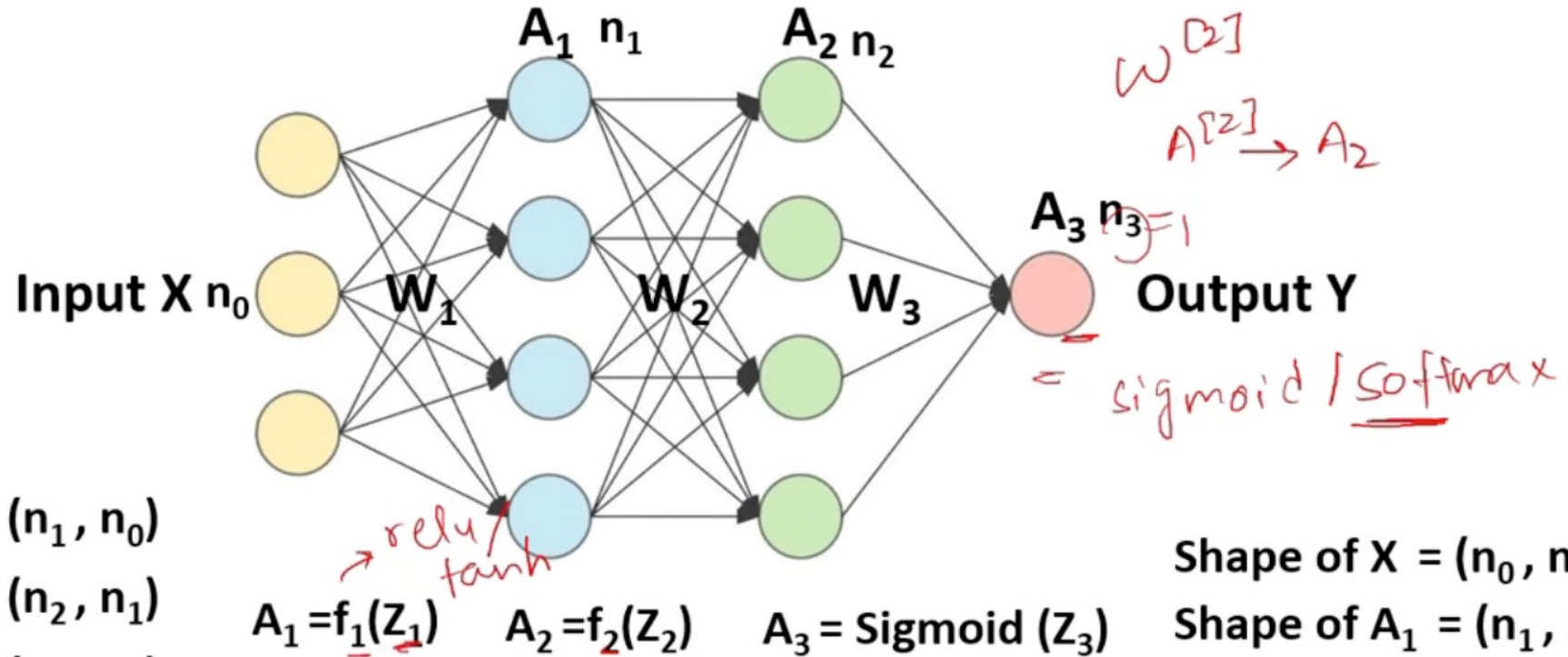


m = total number of observations



Shape of $W_1 = (n_1, n_0)$

Shape of $W_2 = (n_2, n_1)$

Shape of $W_3 = (n_3, n_2)$

Shape of $B_1 = (n_1, 1)$

Shape of $B_2 = (n_2, 1)$

Shape of $B_3 = (n_3, 1)$

$$Z_k = W_k A_{k-1} + B_k$$

$$\text{Cost} = -\frac{1}{m} \sum_{i=1}^m [y_i * \log(a_i) + (1 - y_i) * \log(1 - a_i)]$$

$$\text{Loss}_i = -[y_i * \log(a_i) + (1 - y_i) * \log(1 - a_i)]$$

Shape of $X = (n_0, m)$

Shape of $A_1 = (n_1, m)$

Shape of $A_2 = (n_2, m)$

Shape of $A_3 = (n_3, m)$

Shape of $Y = (n_3, m)$



Forward Pass:

- $Z^{[\ell]} = W^{[\ell]} A^{[\ell-1]} + b^{[\ell]}$
- $A^{[\ell]} = g^{[\ell]}(Z^{[\ell]})$

Backward Pass:

- Loss derivative¹: $dA^{[L]} = \frac{\partial J}{\partial A^{[L]}}$
- $dZ^{[\ell]} = dA^{[\ell]} * g^{[\ell]'}(Z^{[\ell]})$
- $dW^{[\ell]} = \frac{1}{m} dZ^{[\ell]} (A^{[\ell-1]})^T$
- $db^{[\ell]} = \frac{1}{m} np.sum(dZ^{[\ell]}, axis = 1, keepdims = True)$
- $dA^{[\ell-1]} = (W^{[\ell]})^T dZ^{[\ell]}$

Loss Derivative

- Sigmoid + BCE** (binary): $dZ = a - y$.
- Softmax + cross-entropy** (multiclass): $dZ = p - y$.
- Sigmoid + MSE**: $dZ = (a - y) a(1 - a)$
- Tanh/ReLU + MSE/other**: $dZ = dA \odot g'(Z)$.

6.1 - Linear Backward

For layer l , the linear part is: $Z^{[l]} = W^{[l]} A^{[l-1]} + b^{[l]}$ (followed by an activation).

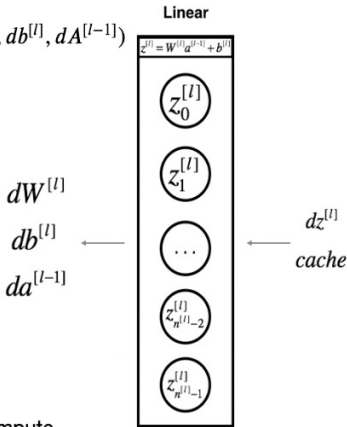
Suppose you have already calculated the derivative $dZ^{[l]} = \frac{\partial \mathcal{L}}{\partial Z^{[l]}}$. You want to get $(dW^{[l]}, db^{[l]}, dA^{[l-1]})$

The three outputs $(dW^{[l]}, db^{[l]}, dA^{[l-1]})$ are computed using the input $dZ^{[l]}$

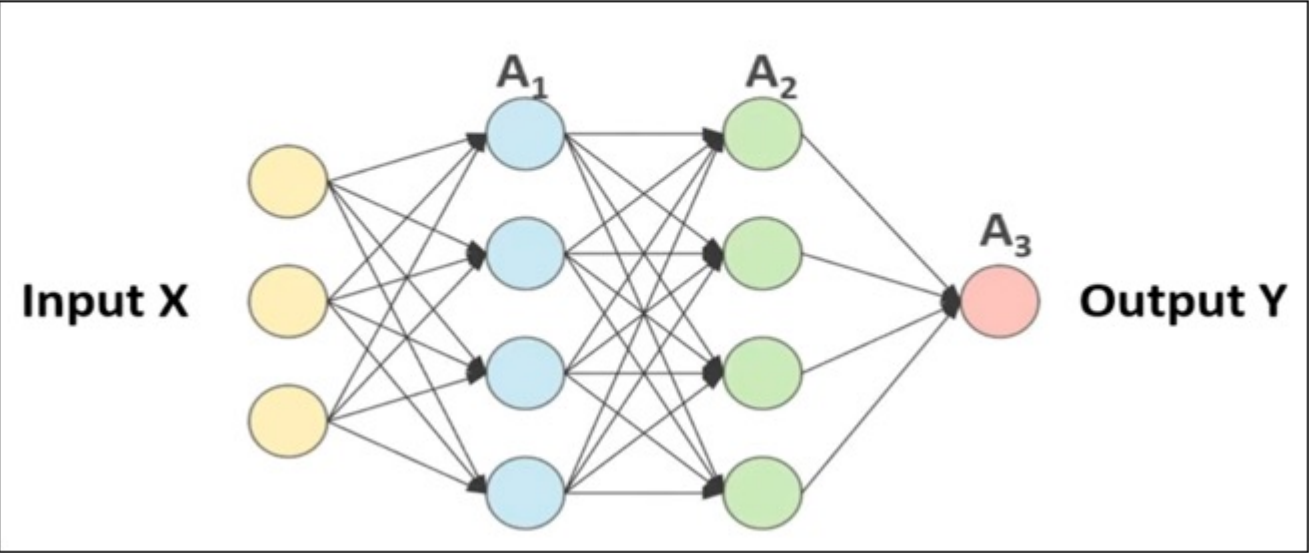
Here are the formulas you need:

$$dW^{[l]} = \frac{\partial \mathcal{J}}{\partial W^{[l]}} = \frac{1}{m} dZ^{[l]} A^{[l-1]T}$$
$$db^{[l]} = \frac{\partial \mathcal{J}}{\partial b^{[l]}} = \frac{1}{m} \sum_{i=1}^m dZ^{[l](i)}$$
$$dA^{[l-1]} = \frac{\partial \mathcal{L}}{\partial A^{[l-1]}} = W^{[l]T} dZ^{[l]}$$

If $g(\cdot)$ is the activation function, `sigmoid_backward` and `relu_backward` compute $dZ^{[l]} = dA^{[l]} * g'(Z^{[l]})$.



m	n0	n1	n2	n3
50	64	32	16	1



X	(n0,m)	(64,50)
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no gradients

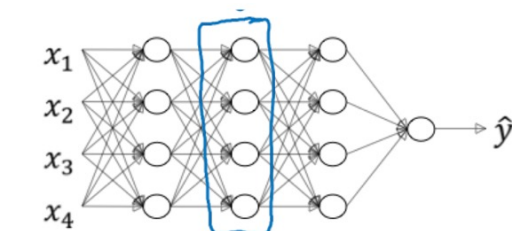
W1,dw1	(n1,n0)	(32,64)
W2,dw2	(n2,n1)	(16,32)
W3,dw3	(n3,n2)	(1,16)

B1,db1	(n1,1)	(32,1)
B2,db2	(n2,1)	(16,1)
B3,db3	(n3,1)	(1,1)

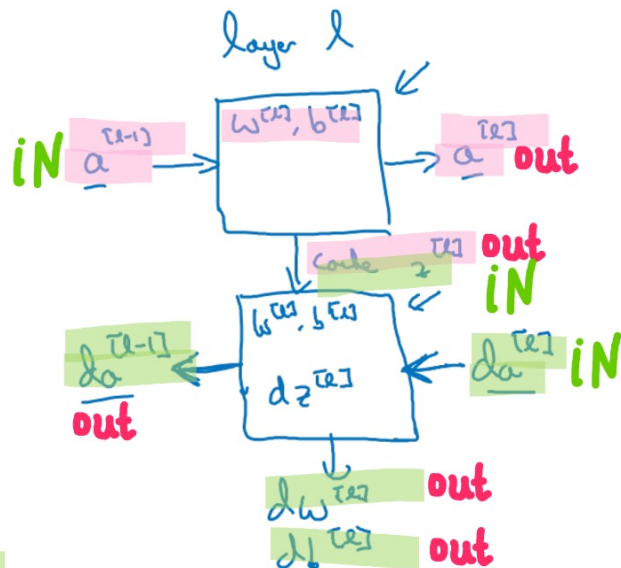
Z1,dz1,A1,da1	(n1,m)	(32,50)	$(n1,n0)@(n0,m)+(n1,1)=(n1,m)$	$(32,64)@(64,50)+(32,1)=(32,50)$
Z2,dz2,A2,da2	(n2,m)	(16,50)	$(n2,n1)@(n1,m)+(n2,1)=(n2,m)$	$(16,32)@(32,50)+(16,1)=(16,50)$
Z3,dz3,A3,da3	(n3,m)	(1,50)	$(n3,n2)@(n2,m)+(n3,1)=(n3,m)$	$(1,16)@(16,50)+(1,1)=(1,50)$

Y	(n3,m)	(1,50)
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no gradients



layer l : $W^{[l]}, b^{[l]}$
 → Forward: Input $a^{[l-1]}$, output $a^{[l]}$
 $z^{[l]} = W^{[l]} a^{[l-1]} + b^{[l]}$ cache $z^{[l]}$
 $a^{[l]} = g^{[l]}(z^{[l]})$
 → Backward: Input $da^{[l]}$, output $da^{[l-1]}$
 cache $(z^{[l]})$
 $\frac{dz^{[l]}}{da^{[l-1]}}$
 $\frac{db^{[l]}}{da^{[l-1]}}$



$$\begin{aligned} Z^{[1]} &= W^{[1]}X + b^{[1]} \\ A^{[1]} &= g^{[1]}(Z^{[1]}) \\ Z^{[2]} &= W^{[2]}A^{[1]} + b^{[2]} \\ A^{[2]} &= g^{[2]}(Z^{[2]}) \\ &\vdots \\ A^{[L]} &= g^{[L]}(Z^{[L]}) = \hat{Y} \end{aligned}$$

$$\begin{aligned} dz^{[l]} &= dA^{[l]} * g'^{[l]}(z^{[l]}) \\ dW^{[l]} &= \frac{1}{m} dz^{[l]} \cdot A^{[l-1]T} \\ db^{[l]} &= \frac{1}{m} np.sum(dz^{[l]}, axis=1, keepdims=True) \\ dA^{[l-1]} &= W^{[l]T} \cdot dz^{[l]} \end{aligned}$$

$$\begin{aligned} dZ^{[L]} &= A^{[L]} - Y \\ dW^{[L]} &= \frac{1}{m} dZ^{[L]} A^{[L-1]T} \\ db^{[L]} &= \frac{1}{m} np.sum(dZ^{[L]}, axis=1, keepdims=True) \\ dZ^{[L-1]} &= W^{[L]T} dZ^{[L]} * g'^{[L-1]}(Z^{[L-1]}) \end{aligned}$$

Note that * denotes element-wise multiplication)

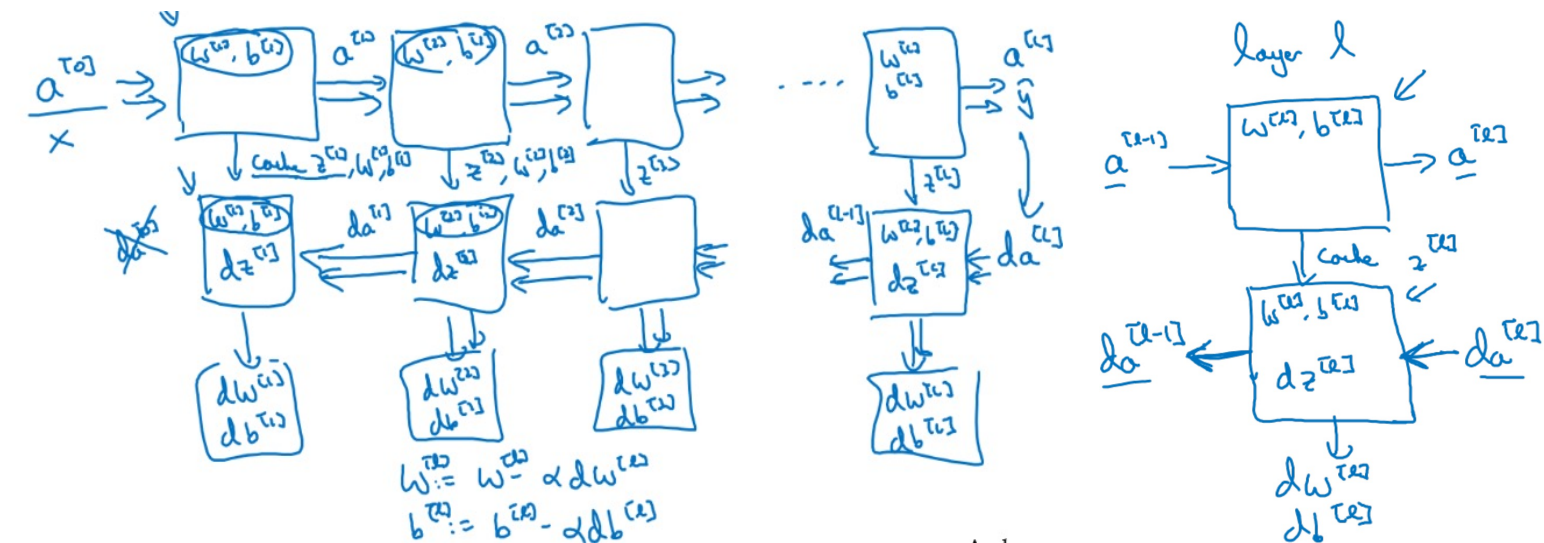
$$dZ^{[1]} = W^{[2]} dZ^{[2]} * g'^{[1]}(Z^{[1]})$$

$$dW^{[1]} = \frac{1}{m} dZ^{[1]} A^{[0]T}$$

Note that $A^{[0]T}$ is another way to denote the input features, which is

$$db^{[1]} = \frac{1}{m} np.sum(dZ^{[1]}, axis=1, keepdims=True)$$

Activation	Formula	Derivative	Loss Function	Formula	Derivative w.r.t. $\hat{y}_i$
Sigmoid	$\sigma(x) = \frac{1}{1+e^{-x}}$	$\sigma'(x) = \sigma(x)(1 - \sigma(x))$	Binary Cross-Entropy (BCE)	$\mathcal{L} = -\frac{1}{m} \sum_{i=1}^m [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)]$	$\frac{\partial \mathcal{L}}{\partial \hat{y}_i} = -\frac{1}{m} \left(\frac{y_i}{\hat{y}_i} - \frac{1 - y_i}{1 - \hat{y}_i} \right)$
Tanh	$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$	$\tanh'(x) = 1 - \tanh^2(x)$			
ReLU	$\text{ReLU}(x) = \max(0, x)$	$\text{ReLU}'(x) = \begin{cases} 1, & x > 0 \\ 0, & x \leq 0 \end{cases}$	Categorical Cross-Entropy (CCE)	$\mathcal{L} = -\frac{1}{m} \sum_{i=1}^m \sum_{k=1}^K y_{ik} \log \hat{y}_{ik}$	$\frac{\partial \mathcal{L}}{\partial \hat{y}_{ik}} = -\frac{1}{m} \frac{y_{ik}}{\hat{y}_{ik}}$
Leaky ReLU	$\text{LeakyReLU}(x) = \begin{cases} x, & x > 0 \\ \alpha x, & x \leq 0 \end{cases}$	$\text{LeakyReLU}'(x) = \begin{cases} 1, & x > 0 \\ \alpha, & x \leq 0 \end{cases}$	Mean Absolute Error (MAE, L1)	$\mathcal{L} = \frac{1}{m} \sum_{i=1}^m y_i - \hat{y}_i $	$y - \hat{y}$
ELU	$\text{ELU}(x) = \begin{cases} x, & x > 0 \\ \alpha(e^x - 1), & x \leq 0 \end{cases}$	$\text{ELU}'(x) = \begin{cases} 1, & x > 0 \\ \alpha e^x, & x \leq 0 \end{cases}$	Mean Squared Error (MSE, L2)	$\mathcal{L} = \frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2$	$\frac{\partial \mathcal{L}}{\partial \hat{y}_i} = \frac{2}{m} (\hat{y}_i - y_i)$
Softplus	$\text{softplus}(x) = \ln(1 + e^x)$	$\text{softplus}'(x) = \sigma(x) = \frac{1}{1+e^{-x}}$ Reddit +15 Wikipedia +10 Wikipedia Medium	MAPE	$(\displaystyle \mathcal{L} = \frac{1}{m} \sum_{i=1}^m \frac{ y_i - \hat{y}_i }{ y_i })$	$\frac{\partial \mathcal{L}}{\partial \hat{y}_{ik}} = -\frac{1}{m} \frac{1[k = y_i]}{\hat{y}_{i, y_i}}$
GELU	$\text{GELU}(x) = x \cdot \Phi(x) = \frac{x}{2} \left[1 + \text{erf}\left(\frac{x}{\sqrt{2}}\right) \right]$	$\text{GELU}'(x) = \Phi(x) + \frac{x}{\sqrt{2\pi}} e^{-x^2/2}$ Wikipedia	Sparse Categorical Cross-Entropy	$\mathcal{L} = -\frac{1}{m} \sum_{i=1}^m \log(\hat{y}_{i, y_i})$	
Softmax	$\text{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$	$\frac{\partial \sigma_i}{\partial z_k} = \sigma_i(\delta_{ik} - \sigma_k)$ Wikipedia			



Forward Pass:

- $Z^{[l]} = W^{[l]}A^{[l-1]} + b^{[l]}$
- $A^{[l]} = g^{[l]}(Z^{[l]})$

Backward Pass:

- Loss derivative¹: $dA^{[L]} = \frac{\partial J}{\partial A^{[L]}}$
- $dZ^{[l]} = dA^{[l]} * g^{[l]'}(Z^{[l]})$
- $dW^{[l]} = \frac{1}{m} dZ^{[l]} (A^{[l-1]})^T$
- $db^{[l]} = \frac{1}{m} np.sum(dZ^{[l]}, axis = 1, keepdims = True)$
- $dA^{[l-1]} = (W^{[l]})^T dZ^{[l]}$

